CAMPUSLINK

UCS 503 • SOFTWARE ENGINEERING



CampusLink

“Find your people — beyond your friend circle.”

A goal-matching and resource-sharing platform for college campuses

SOFTWARE REQUIREMENTS SPECIFICATION

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Title	CampusLink — Software Requirements Specification
Course	UCS503 Software Engineering, Computer Science and Engineering Department
Institute	Thapar Institute of Engineering and Technology, Patiala
Project repository	`github.com/govinda16293/CampusLink`
Status	Prepared for evaluation. Build status is recorded per requirement (Chapter 3).
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Project at a glance

15	66	16	4 of 13
use cases	numbered requirements	REST endpoints live	build steps complete

How to read the status markers

- IMPL **Implemented.** Built, merged in the repository and covered by automated tests or a screen in the prototype.
- PART **Partial.** Part of the behaviour or its data model exists; the rest waits for a later build step.
- PLAN **Planned.** Specified and scheduled in the build order of Section 5.3; not yet in the repository.
- DESC **Descoped.** Considered in the proposal and deliberately cut. Kept here so the decision is on record.

> This specification keeps the *specified* product and the *built* product apart. Of the 66 requirements, 29 are implemented, 4 are partial, 31 are planned and 2 were descoped. Nothing marked PLAN is claimed to exist in the prototype.

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1 Introduction

1.1 Purpose

On a campus of several thousand students, everyday coordination still runs through batch WhatsApp groups, hostel groups and word of mouth: a gym partner, a shared cab to the airport, someone who has last semester's notes. That works only if you already know the right people. It also gives no safe structure for dealing with a stranger, and no record of who keeps flaking on plans.

CampusLink is a web platform that closes this gap. A student *posts a goal* (an activity that needs another person). The goal appears on a public campus feed. Interested students *request to join*. Once enough are in, a small *lobby* forms and the group takes it from there. A trust-and-safety layer gives the interaction enough structure to be safe with someone you do not know yet: anonymous posting, a gender-preference filter, a visible reliability score, report and block, and no-show tracking.

This document is the Software Requirements Specification (SRS) for CampusLink. It states what the system shall do, the constraints it works under and the quality it must reach. For every requirement it also records whether the requirement is already built, partly built, planned or descoped, so the document can be read against the repository now and again at final submission.

1.2 Project Scope

CampusLink serves one closed community: students holding a verified `@thapar.edu` address. Within it, the product supports the loop *post, discover, request, approve, lobby, coordinate, close and rate*. The same loop is reused for a smaller second use case, academic resource sharing: a student posts that they need notes for a subject, and whoever has them answers in a private one-to-one thread instead of a public lobby.

The success of the product is judged by one primary metric, the *lobby fulfilment rate* (the share of posted goals that reach their required headcount in time), and by four secondary metrics defined in Section 4.4. They will be measured in a pilot with 15–20 students from the team's own batch.

In scope

- College-domain accounts with e-mail OTP
- Goal posting, public feed, filters
- Request-to-join with manual or automatic approval
- Lobbies and real-time chat
- Notes and resource requests with private threads
- Anonymous posting, gender preference, report and block
- Reliability score, attendance confirmation, ratings
- Rule-based suggestions and priority ordering
- Evaluation metrics and free-tier deployment

1.3 System Work-flow Overview

Figure 1.1 shows the core loop in seven stages. The first two stages are built and tested. The trust-and-safety layer is not a stage of its own: it wraps every stage, because its job is to make each hand-off between strangers safer.

Every stage depends on the one before it. The poster's goal starts the cycle; the feed brings it to people who want the same thing; the request and approval step keeps the poster in control of who joins; the lobby turns accepted strangers into a group; and the closing step feeds back into the reliability score, which every later feed card displays.

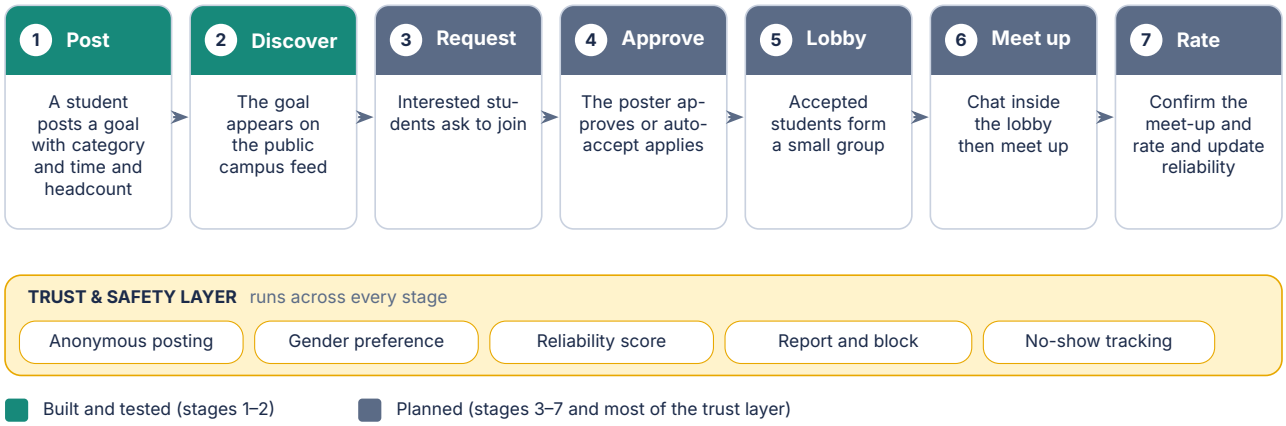


Figure 1.1. Entire system work-flow: the seven stages of the CampusLink loop and the trust layer that spans them

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
Goal	An activity a student needs other people for: a category, a title, a description, an optional date and time and a headcount.
Poster	The student who posted a goal.
Requester	A student who asked to join someone else's goal.
Headcount	The number of *other* people the poster needs. The poster is not counted; a gym partner is a headcount of 1.
Feed	The public list of open goals, newest first, filterable by category and date.
Lobby	The group formed around a goal once its first request is accepted. Members chat inside it.
Anonymous goal	A goal whose poster's name and photo are hidden from the feed until a requester is accepted. The reliability score stays visible.
Gender preference	A per-goal option (`ANY`, `MALE_ONLY`, `FEMALE_ONLY`) restricting who may request to join.
Auto-accept	A per-goal option that accepts valid requests immediately until the headcount is met.
Reliability score	An integer, starting at 100, that summarises how dependable a student has been. Shown on every profile and feed card.
OTP	One-time passcode: six digits, e-mailed at sign-up, valid for 10 minutes.
JWT	JSON Web Token carried in the `Authorization: Bearer` header after login.
DTO	Data transfer object: the JSON shape the API returns, built by a serializer from a database row.
SPA	Single-page application. The React front end.
ORM	Object-relational mapper. Prisma in this project.
Zod	TypeScript schema-validation library, used for request validation and shared between front end and back end.
scrypt	Memory-hard password-hashing function used for stored passwords.
UC, FR, NFR	Use case, functional requirement, non-functional requirement.

1.5 References

[1] CampusLink Project Proposal, UCS503, August 2026.

[2] CampusLink Project Report, prototype stage, UCS503, September 2026.

[3] CampusLink source repository, `github.com/govinda16293/CampusLink` (backend, frontend and shared packages; journals and design notes).

[4] ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.

[5] Course template: Software Requirements Specification (UCS503), and the *Property Pulse* SRS used as the format reference.

2 Overall Description

2.1 Product Perspective

CampusLink is a new, self-contained product; it replaces nothing that exists in software, only informal chat groups. It is a 3-tier web application: a React single-page application in the browser, a stateless Express API, and a relational database reached only through the API. The only systems it talks to are an SMTP relay for the sign-up passcode and, later, an object store for notes attachments. Figure 2.1 draws that boundary, and Figure 2.2 opens the box.

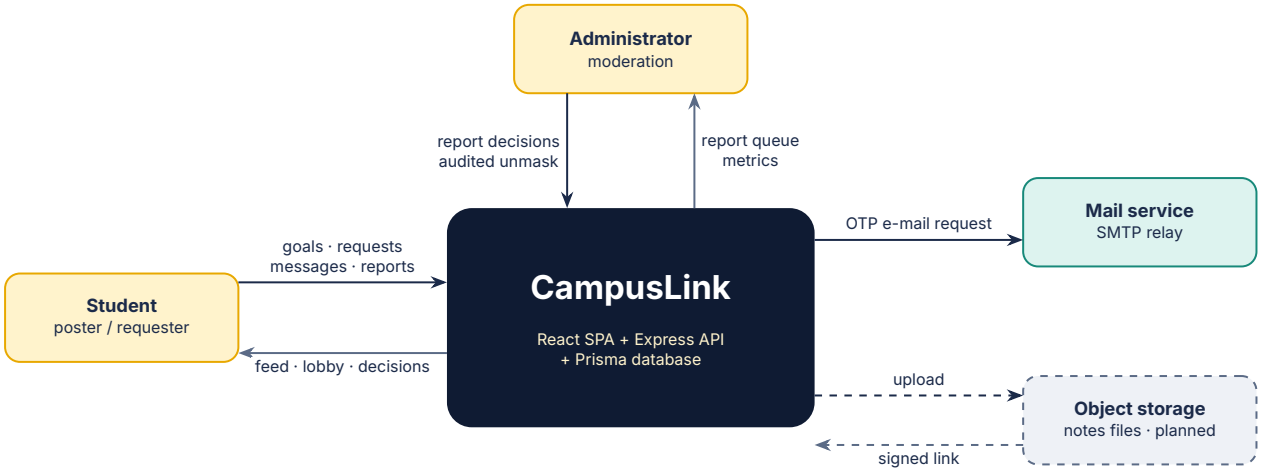


Figure 2.1. System context: CampusLink and the people and services around it

2.2 User Classes and Characteristics

The system distinguishes three user classes . Within a goal, one student account plays one of three roles at a time.

- **Visitor:** not signed in. Sees the landing page and can sign up, verify the e-mail OTP and log in. Cannot read the feed.
- **Student:** a verified `@thapar.edu` account. Posts goals, browses the feed, requests to join, chats in lobbies, rates, reports and blocks. Comfortable with a phone browser; expects pages to load quickly.
- **Administrator:** a student account whose `role` is `ADMIN`. Reviews reports, may unmask an anonymous poster for a recorded reason, and reads the evaluation metrics.

2.3 Product Functions

The core product functions are:

1. **Verified, college-only accounts.** Sign-up restricted to the college domain and confirmed by an e-mailed OTP.
2. **Goal posting and public feed.** Post an activity, filter the feed by category and date, and read a goal in detail.

3. **Request and approval.** Ask to join; the poster approves, rejects, or lets auto-accept decide.
4. **Lobby and real-time chat.** A group chat opens once a lobby forms.
5. **Notes and resource sharing.** Private one-to-one answers to a notes request, with file attachments.
6. **Trust and safety.** Anonymous posting, gender preference, report and block, and audited administrator unmasking.
7. **Reliability and feedback.** Attendance confirmation, no-show flags, ratings and a visible reliability score.
8. **Matching and metrics.** Rule-based suggestions, reliability-aware request ordering and the evaluation metrics of Section 4.4.

2.4 Operating Environment

CampusLink runs in any standard computing environment that has a modern web browser on the client side and Node.js on the server side.

Layer	Technology
Client	Chrome, Firefox, Safari, Edge and mobile browsers. React 18, Vite 6, Tailwind CSS v4, React Router 7, Zustand 5; plain JavaScript.
Server	Node.js and Express 5 in TypeScript; Zod 3 validation; JWT sessions; `scrypt` password hashing; nodemailer for SMTP; helmet and `express-rate-limit`.
Data	Prisma 5 ORM over SQLite in development and PostgreSQL in production.
Shared code	A workspace package of Zod schemas, enumerations and constants imported by both front end and back end.
Quality tooling	Vitest and Supertest for tests; ESLint, Prettier and the TypeScript compiler; GitHub Actions runs format, lint, type-check, test and build on every push.
Planned	Socket.IO for chat, Cloudinary or Firebase for files; Vercel for the front end, Render or Railway for the API.

Table 2.1. Operating environment and technology stack

2.5 Design and Implementation Constraints

- The platform shall be usable on an average student phone or laptop, with no heavy client-side processing.
- The deployment shall fit standard free-tier hosting, so that a live demonstration link exists.
- Free-tier hosts have an ephemeral file system. Avatars are therefore stored in the database until object storage arrives in Step 7.
- SQLite has no native enumerations or arrays. Enumerated columns are strings checked at the API boundary, and interests are stored as one comma-separated string.
- Identity is guaranteed only by the college e-mail domain; there is no other identity proof.

2.6 Assumptions and Dependencies

- Students have a working `@thapar.edu` mailbox and can receive e-mail within a few minutes.
- The SMTP relay is available. If it is not, sign-up cannot complete, because verification is the only way to activate an account.
- The pilot has 15 to 20 users, so the feed is small and no sharding or caching layer is needed.
- Each goal's date and time are interpreted in the poster's browser time zone and stored in UTC.
- Verification happens once, at sign-up. Later logins use the password only.

2.7 System Architecture

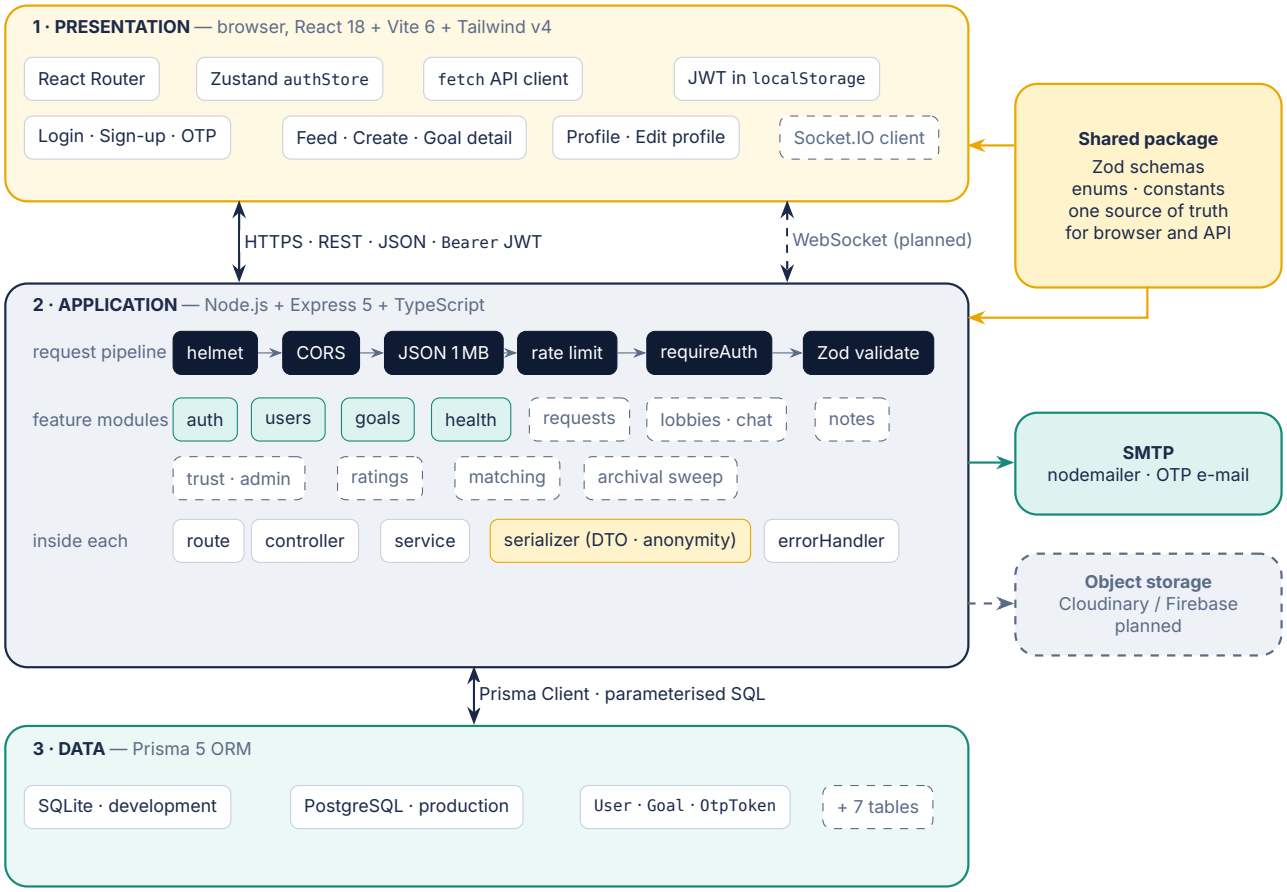


Figure 2.2. Three-tier architecture of CampusLink; dashed items are planned

Figure 2.2 shows the three tiers and the pipeline every request passes through. Business logic sits in services; a serializer is the only place that turns a database row into a response, which is why anonymity is enforced there and never in the browser. Dashed items are planned.

2.8 Design Activities and Architectural Models

2.8.1 System UML Use Case Diagram

Figure 2.3 specifies the interactions between the actors and the system. The mail service delivers the OTP for *Verify E-mail OTP*; *Sign Up* and *Report or Block User* `include` the verification and review use cases, and *Coordinate in Lobby Chat* follows *Request to Join* through an `extend` relationship that applies only if the request is accepted. The catalogue beside the diagram gives each use case an identifier, which the requirement tables in Chapter 3 trace back to.

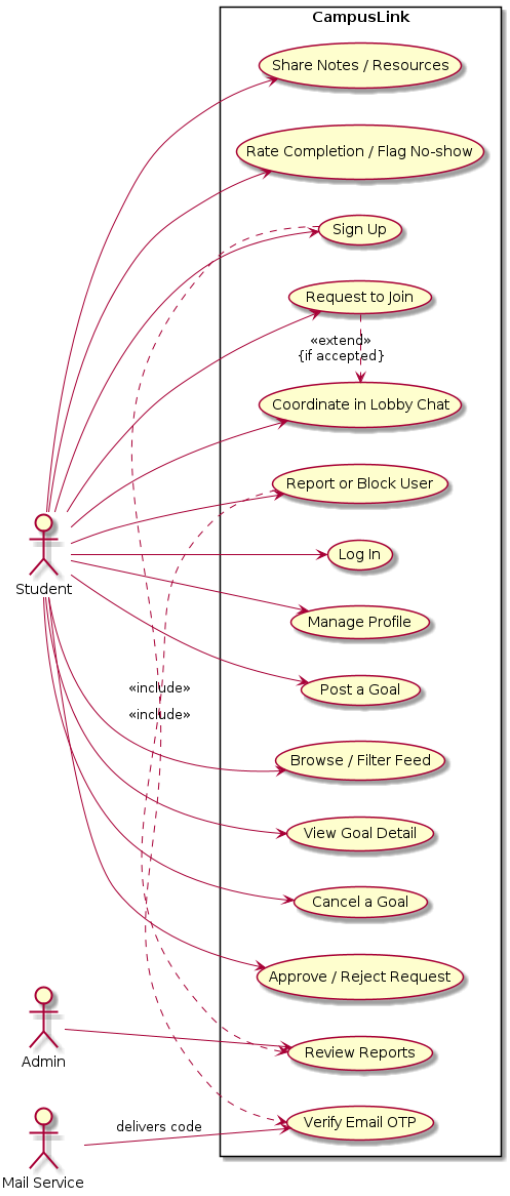
2.8.2 Activity Diagram

Figure 2.4 follows one student through the whole product: sign-up and verification, posting or browsing, the request and approval decision, the lobby, and the closing step with confirmation, rating and no-show flagging.

2.8.3 UML Class Diagram and Relational Schema

Figure 2.5 defines the domain model. Figure 2.6 shows the three tables that exist in the database today (`User`, `Goal`, `OtpToken`); seven more arrive with Steps 4 to 9, as drawn in the class diagram.

The link between `OtpToken` and `User` is by e-mail address only and has no foreign key. That is deliberate: a passcode is issued before the account is verified, and a code row must be able to outlive nothing but its own



ID	Use case	Actor(s)	Status
UC-01	Sign Up	Student	IMPL
UC-02	Verify E-mail OTP	Student, Mail Service	IMPL
UC-03	Log In	Student	IMPL
UC-04	Manage Profile	Student	IMPL
UC-05	Post a Goal	Student	IMPL
UC-06	Browse / Filter Feed	Student	IMPL
UC-07	View Goal Detail	Student	IMPL
UC-08	Cancel a Goal	Student	IMPL
UC-09	Request to Join	Student	PLAN
UC-10	Approve / Reject Request	Student (poster)	PLAN
UC-11	Coordinate in Lobby Chat	Student	PLAN
UC-12	Share Notes / Resources	Student	PLAN
UC-13	Rate Completion / Flag No-show	Student	PLAN
UC-14	Report or Block User	Student	PLAN
UC-15	Review Reports	Administrator	PLAN

Figure 2.3. System use case diagram and the use-case catalogue (IDs used throughout this SRS)

expiry. Foreign keys with cascade delete are used elsewhere; deleting a student removes their goals.

2.8.4 Sequence Diagram: Browse, Request and Approval

Figure 2.7 shows the chronological exchange when a requester opens the feed, asks to join and is accepted. The front end asks the back end for goals, the back end reads them from the database, the request is forwarded to the poster, and the approval flows back until the requester lands in the lobby. The first half (feed) is built; the request, notification and lobby messages are planned for Steps 4 and 5.

2.8.5 Sequence Diagram: Sign-up and OTP Verification

Figure 2.8 shows the sign-up path, which is built and tested. The API validates the college domain, stores a salted `scrypt` hash of the password and only a hash of the passcode, and e-mails the code. A wrong code increments an attempt counter and the code is burned after five misses, so a six-digit code cannot be brute-forced inside its ten-minute life.

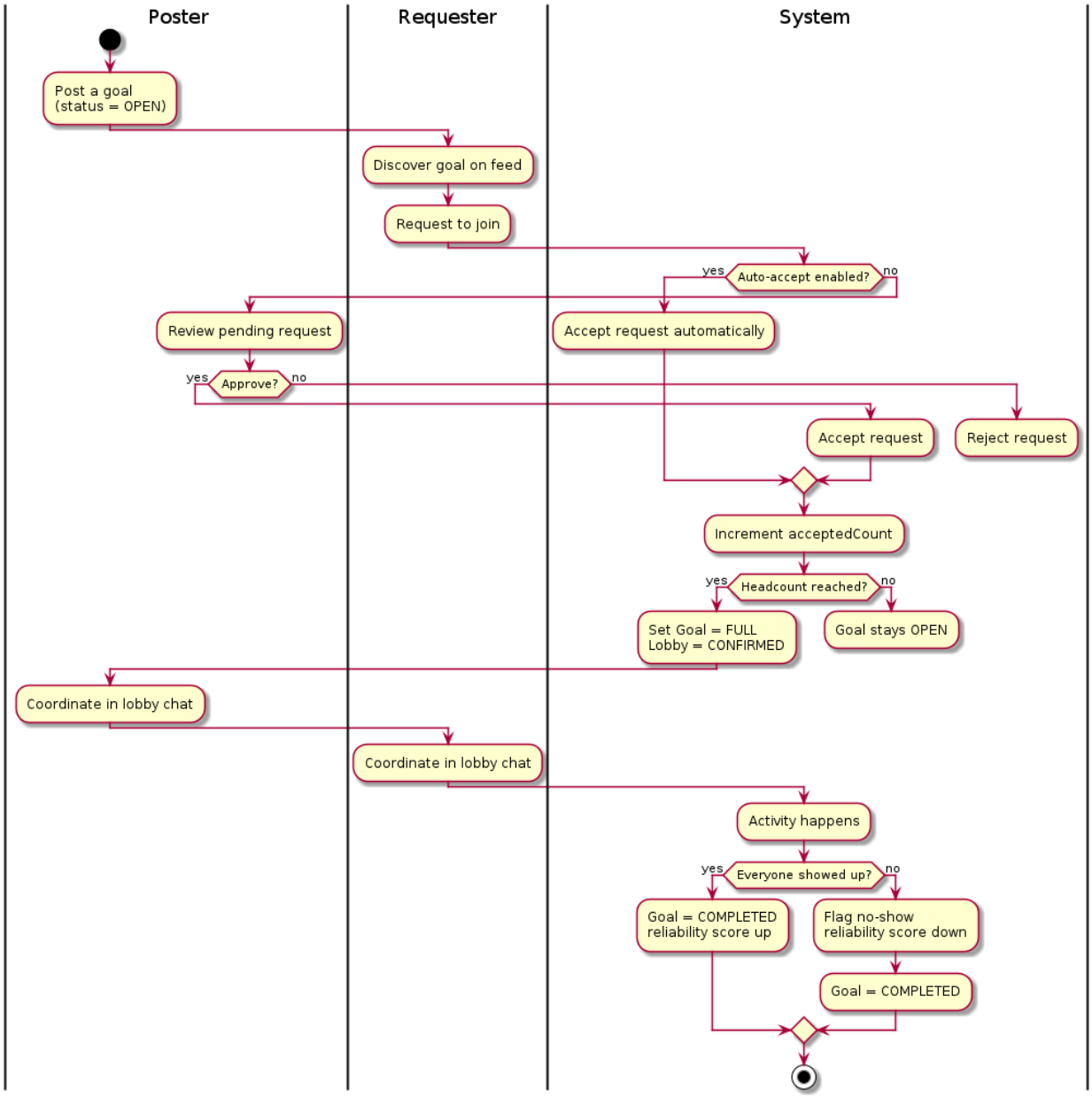


Figure 2.4. Activity diagram of the CampusLink lifecycle

2.8.6 Data-flow Diagrams

Figure 2.9 covers the three built processes; the remaining processes (requests, lobby chat, trust and safety, ratings) follow the same pattern and are planned. Each arrow is labelled F1, F2 and so on. Round boxes are processes, open rectangles are data stores and shaded boxes are external entities.

2.8.7 State-machine Models

The goal is the main entity with a life cycle (Figure 2.10). Solid transitions are implemented; dashed transitions are planned. Join requests move through `PENDING`, `ACCEPTED`, `REJECTED` or `WITHDRAWN`, and lobbies through `FORMING`, `CONFIRMED`, `COMPLETED` or `CANCELLED` (Steps 4 and 5).

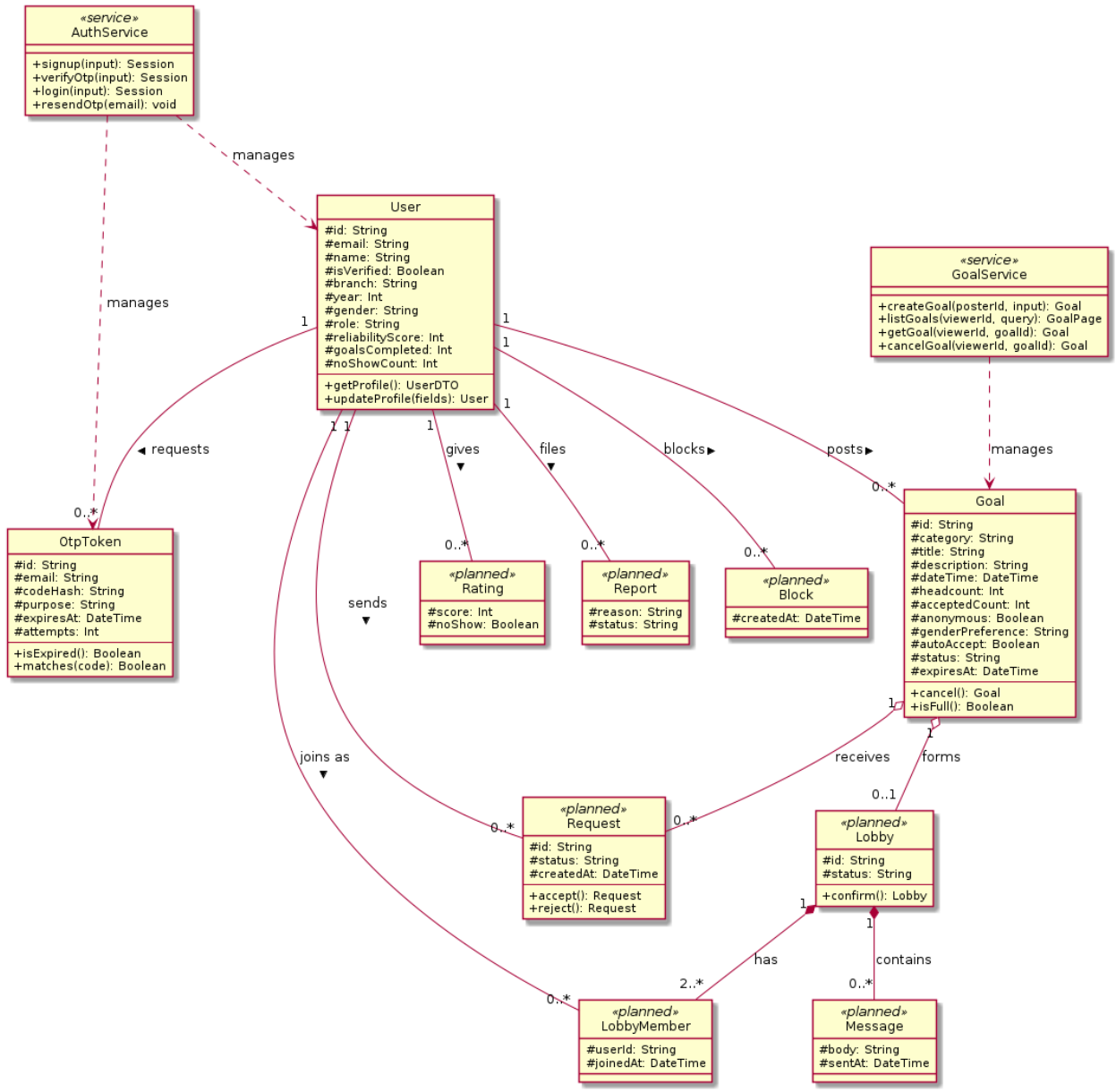


Figure 2.5. UML class diagram of the CampusLink domain

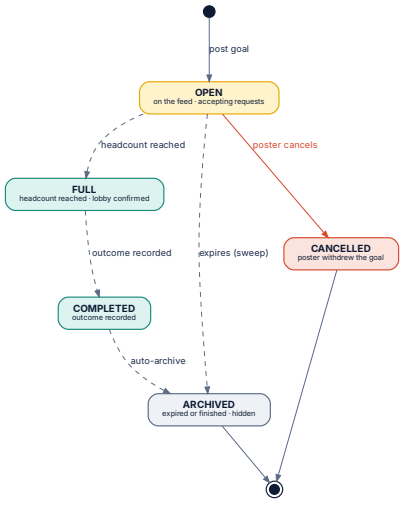


Figure 2.10. State machine of a goal

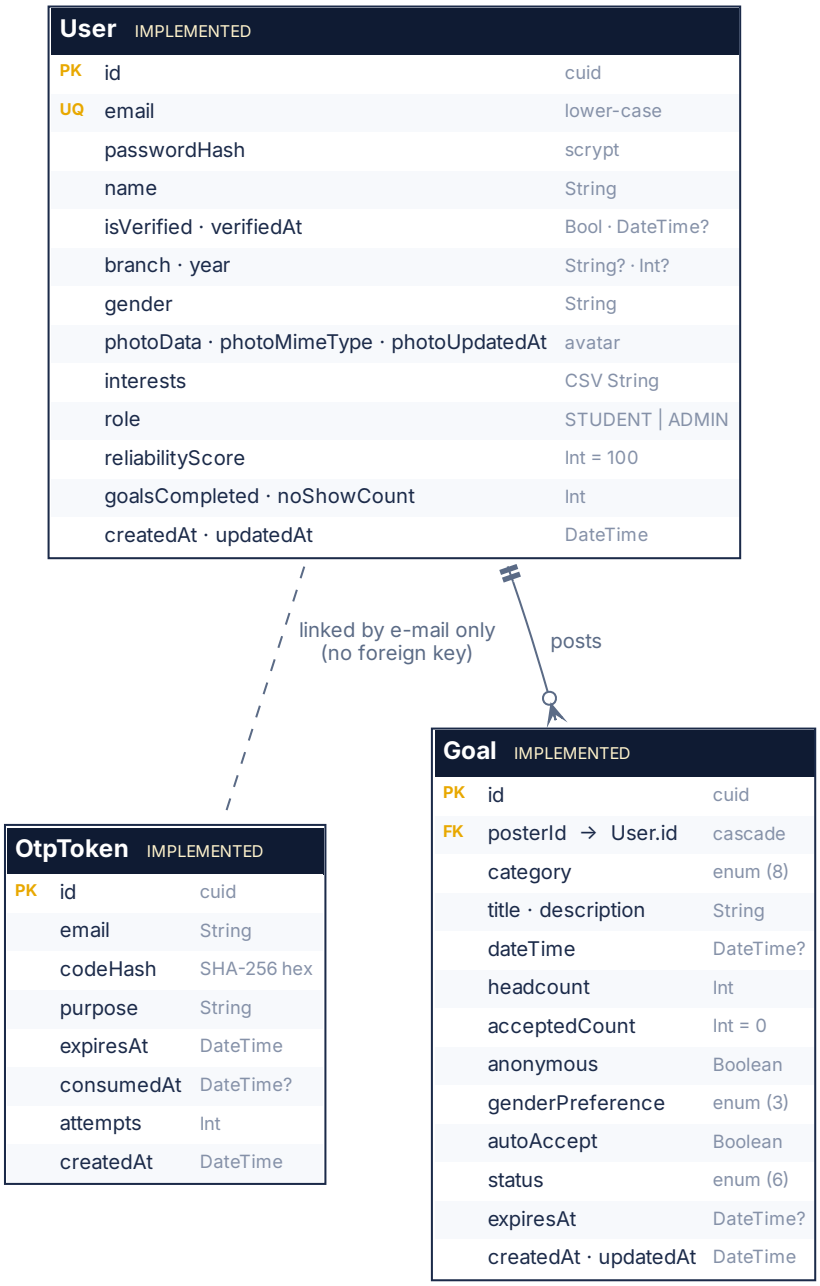


Figure 2.6. Relational schema, implemented part (`User`, `Goal`, `OtpToken`)

From	To	Trigger	Status
(start)	**OPEN**	The poster submits a valid goal.	IMPL
OPEN	**CANCELLED**	The poster cancels. Only an open goal can be cancelled; anyone else gets `403`.	IMPL
OPEN	**FULL**	The accepted count reaches the headcount.	PLAN
FULL	**COMPLETED**	Outcomes of the meet-up are recorded.	PLAN
OPEN	**ARCHIVED**	The goal expires and the sweep marks it. Until then expired goals are hidden at query time.	PLAN
COMPLETED	**ARCHIVED**	Automatic archival after completion.	PLAN

Table 2.2. Goal transitions

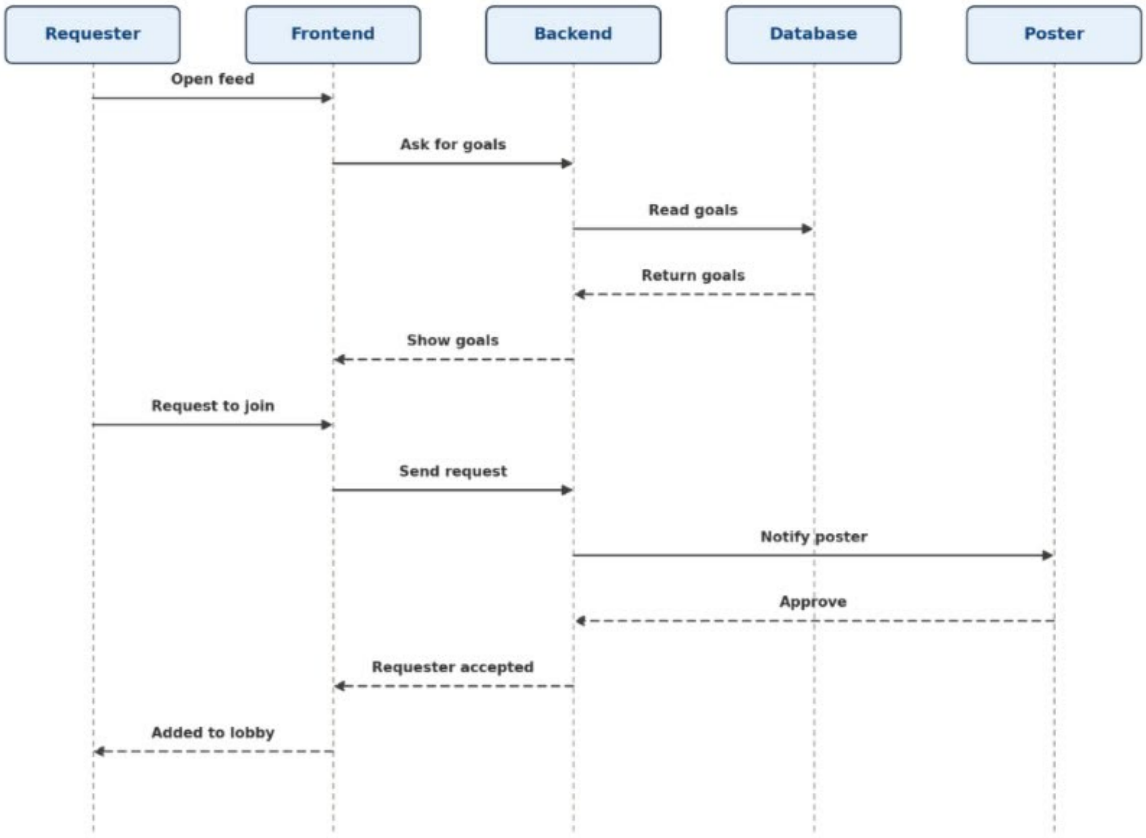


Figure 2.7. Sequence diagram: feed browsing, request to join, approval and entry to the lobby

The enumeration also contains `CLOSED`. No transition into it is defined yet, so it is reserved and unused.

3 System Features

3.1 Description and Priority

Features are grouped into three priority tiers. P1 features form the core loop; without them the product does not work. P2 features make it trustworthy. P3 features improve discovery and measurement.

ID	Prio.	Feature	Description	Module	Status
FEAT-01	High	**Verified student accounts**	Sign-up restricted to the college domain, e-mail OTP, JWT sessions.	M1	IMPL
FEAT-02	High	**Goal posting and feed**	Post a goal; filter and page through the public feed; goal detail; cancel.	M3	IMPL
FEAT-03	High	**Join request and approval**	Request to join; manual approval or auto-accept; gender-preference enforcement.	M4	PLAN
FEAT-04	High	**Lobby and real-time chat**	Group formed from accepted requests; Socket.IO chat with history.	M5	PLAN

ID	Prio.	Feature	Description	Module	Status
FEAT-05	High	**Trust and safety layer**	Anonymous posting, gender preference, report and block, audited unmasking.	M3, M7	PART
FEAT-06	Med	**Student profile**	Branch, year, interests, avatar, completeness, public view.	M2	IMPL
FEAT-07	Med	**Reliability and ratings**	Attendance confirmation, no-show flags, ratings, reliability score.	M8	PART
FEAT-08	Med	**Notes and resource sharing**	Private one-to-one threads and file attachments for notes requests.	M6	PLAN
FEAT-09	Low	**Matching and priority queue**	Suggested goals and reliability-aware ordering of requests.	M9	PLAN
FEAT-10	Low	**Metrics, archival, deployment**	Evaluation metrics, archival sweep, CI, free-tier deployment.	M10	PART

3.2 Functional Requirements (Module Breakdown)

The functional requirements are split into ten modules that follow the build order. Each table lists the requirement, its priority and its status; use-case traceability is in the catalogue of Section 2.8. Figure 3.1 summarises the coverage.

3.2.1 Module 1: Authentication and Account Verification

Build Step 1 11 of 12 implemented

College-domain sign-up, e-mail OTP verification and session handling.

ID	Requirement	Pri.	Status
FR-1.1	The system shall register a student from a name, an e-mail address and a password. The e-mail shall be stored lower-cased and its domain shall be `thapar.edu`; any other domain shall be rejected.	High	IMPL
FR-1.2	The password shall be 8 to 128 characters long and shall be stored only as a salted `scrypt` hash.	High	IMPL
FR-1.3	On sign-up the system shall e-mail a 6-digit one-time passcode (OTP). Only a hash of the code shall be stored; the code shall expire after 10 minutes and be single-use.	High	IMPL
FR-1.4	A code shall be burned after 5 incorrect attempts, after which the student shall have to request a new one.	High	IMPL
FR-1.5	A student shall be able to request a new code. The response shall be identical for unknown and already-verified addresses, so that the endpoint cannot be used to find out who has signed up.	Med	IMPL
FR-1.6	Successful verification shall mark the account as verified and return a session token together with the private user record.	High	IMPL
FR-1.7	Login with e-mail and password shall return a signed JWT (default lifetime 7 days) and the private user record. The error for an unknown e-mail and for a wrong password shall be the same.	High	IMPL
FR-1.8	A student whose account is not yet verified shall not be able to log in and shall be told to complete verification.	High	IMPL
FR-1.9	Every route other than health, the auth endpoints and the avatar image shall require an `Authorization: Bearer` token and shall answer `401` when it is missing, malformed or expired.	High	IMPL
FR-1.10	The authentication endpoints shall be rate-limited to 20 requests per client per 15 minutes.	Med	IMPL
FR-1.11	The client shall let a student log out by discarding the stored session token.	Low	IMPL
FR-1.12	A student who has forgotten the password shall be able to reset it through an e-mailed OTP. The login page already shows a "Forgot password?" link; no reset endpoint exists yet.	Med	PLAN

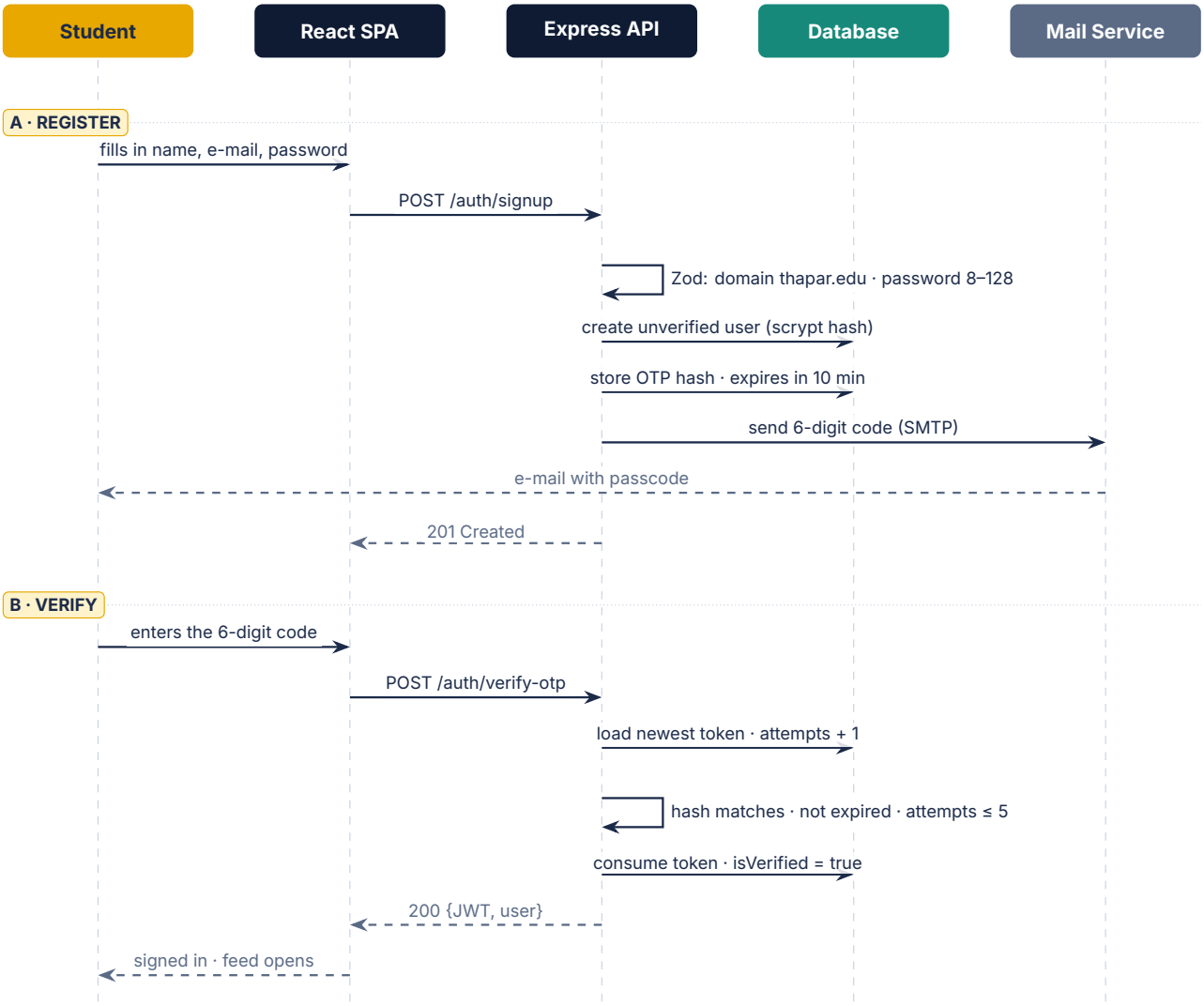


Figure 2.8. Sequence diagram: registration and e-mail OTP verification

3.2.2 Module 2: Student Profile

Build Step 2 — 7 of 7 implemented

Profile fields, avatar, completeness and the public view of another student.

ID	Requirement	Pri.	Status
FR-2.1	An authenticated student shall be able to read their own profile, including e-mail and a profile-completeness percentage.	High	IMPL
FR-2.2	A student shall be able to edit name (2 to 80 letters, spaces, hyphens, apostrophes and periods), branch, year of study (1 to 4) and gender (`MALE`, `FEMALE`, `OTHER`, `UNDISCLOSED`).	High	IMPL
FR-2.3	A student shall be able to keep up to 10 interest tags, each at most 24 characters and free of commas. Case-insensitive duplicates shall be dropped.	Med	IMPL
FR-2.4	The profile update shall reject unknown fields. A student shall not be able to set their own role, verification status, reliability score or counters.	High	IMPL
FR-2.5	A student shall be able to upload and delete an avatar (JPEG, PNG or WebP, resized to 320 px, at most 250 KB).	Low	IMPL
FR-2.6	Avatar bytes shall be served from their own cacheable endpoint and shall never be inlined into goal or user JSON.	Low	IMPL
FR-2.7	A student shall be able to view another student's public profile (name, branch, year, interests, reliability score, goals completed). The e-mail address and role shall never be included.	Med	IMPL

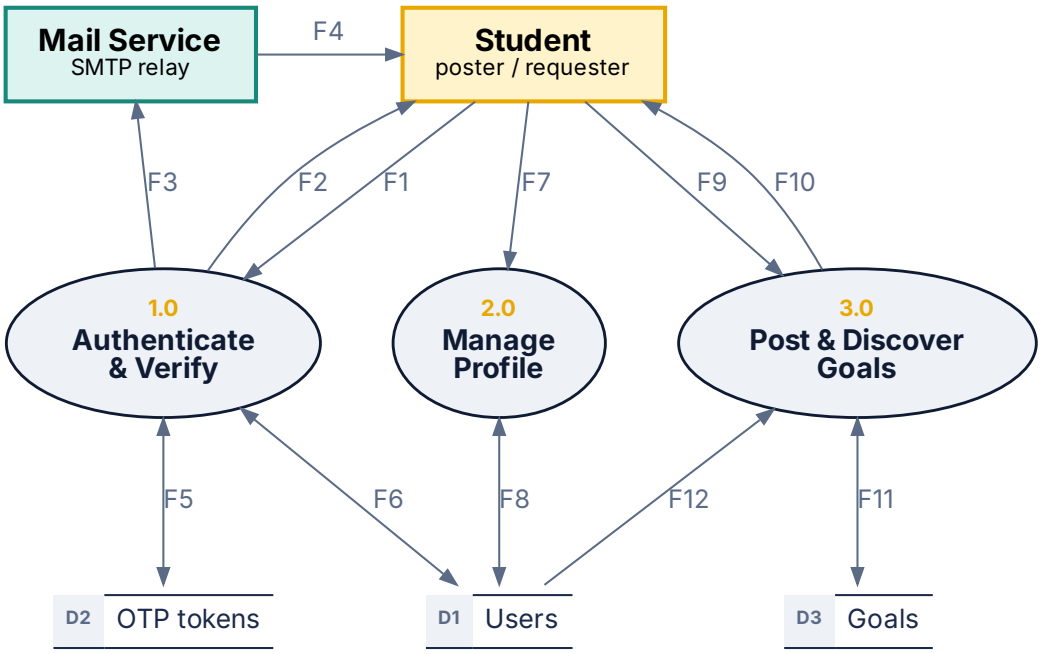


Figure 2.9. Data-flow diagram, implemented processes (1.0 to 3.0)

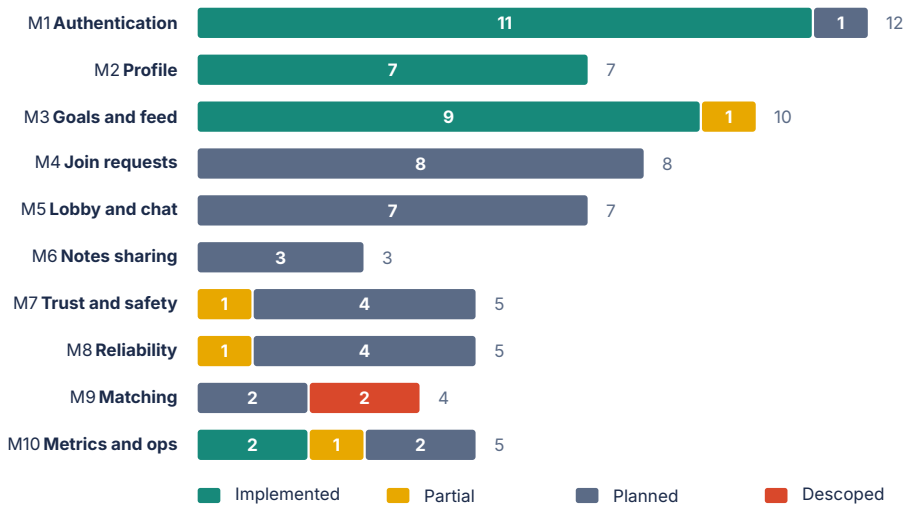


Figure 3.1. Requirements per module by build status (number of requirements at the right of each bar)

3.2.3 Module 3: Goal Posting, Feed and Discovery

Build Step 3 — 9 of 10 implemented

Posting a goal, the campus feed, anonymity and cancellation.

ID	Requirement	Pri.	Status
FR-3.1	A verified student shall be able to post a goal with a category (`SPORTS`, `GYM`, `STUDY`, `TRAVEL`, `FOOD`, `EVENTS`, `NOTES`, `OTHER`), a title of 4 to 80 characters, a description of 10 to 600 characters, a headcount of 1 to 20 (not counting the poster) and an optional date and time.	High	IMPL

ID	Requirement	Pri.	Status
FR-3.2	A goal shall carry three trust options: anonymous posting, a gender preference (`ANY`, `MALE_ONLY`, `FEMALE_ONLY`) and auto-accept.	High	IMPL
FR-3.3	A dated goal shall not be accepted for a time in the past (one minute of clock tolerance is allowed).	Med	IMPL
FR-3.4	A dated goal shall expire at its date and time. An undated goal shall expire 48 hours after it is created.	High	IMPL
FR-3.5	The feed shall list only open, unexpired goals, newest first, with cursor pagination (default page size 20, maximum 50).	High	IMPL
FR-3.6	The feed shall be filterable by category and by date range, and shall offer a switch that shows only the viewer's own goals.	High	IMPL
FR-3.7	For an anonymous goal the server shall withhold the poster's name and photo from everyone except the poster. The poster's reliability score shall remain visible. The rule shall be applied in the server-side serializer, never in the UI.	High	IMPL
FR-3.8	An accepted requester shall be able to see the identity of an anonymous poster. The serializer branch exists; it takes effect once join requests are built.	High	PART
FR-3.9	A student shall be able to open the detail view of a single goal.	High	IMPL
FR-3.10	A poster shall be able to cancel their own open goal (`CANCELLED`). Another student shall receive `403`; a goal that is no longer open shall receive `409`.	Med	IMPL

3.2.4 Module 4: Join Requests and Approval

Build Step 4 0 of 8 implemented

Request-to-join, manual approval or auto-accept, and gender-preference enforcement.

ID	Requirement	Pri.	Status
FR-4.1	A verified student shall be able to request to join an open, unexpired goal that is not their own. At most one active request per student per goal shall exist.	High	PLAN
FR-4.2	A request shall be refused when the goal is not `OPEN`, has expired or is full, or when either student has blocked the other.	High	PLAN
FR-4.3	The gender preference shall be enforced on the server: for `MALE_ONLY` or `FEMALE_ONLY` only a requester whose declared gender matches may request. A requester with gender `UNDISCLOSED` or `OTHER` shall be refused.	High	PLAN
FR-4.4	The poster shall see the pending requests with the requester's identity (never anonymised) and shall be able to approve or reject each one.	High	PLAN
FR-4.5	When the goal has auto-accept set, a valid request shall be accepted immediately until the headcount is met.	Med	PLAN
FR-4.6	Accepting a request shall increase the goal's accepted count. When the count equals the headcount the goal shall move to `FULL` and further requests shall be refused.	High	PLAN
FR-4.7	A requester shall be able to withdraw a pending request (`WITHDRAWN`).	Med	PLAN
FR-4.8	The poster shall be notified of a new request and the requester of the decision.	Med	PLAN

ID	Requirement	Pri.	Status

3.2.5 Module 5: Lobby and Real-time Chat

Build Steps 5 and 6 0 of 7 implemented

Group lobby formed from accepted requests and the chat inside it.

ID	Requirement	Pri.	Status
FR-5.1	When the first request for a goal is accepted a lobby shall be created in state `FORMING`, with the poster and every accepted student as members.	High	PLAN
FR-5.2	The lobby shall move to `CONFIRMED` when the headcount is reached, to `COMPLETED` when the outcome is recorded, and to `CANCELLED` if the goal is cancelled while the lobby is still forming.	High	PLAN
FR-5.3	Only lobby members shall be able to read or post messages in the lobby; any other student shall receive `403`.	High	PLAN
FR-5.4	Messages shall be delivered in real time over WebSocket (Socket.IO). They shall be persisted, and the history shall be loaded when a member opens the lobby.	High	PLAN
FR-5.5	A message shall be limited in length (proposed: 1000 characters) and shall be rendered as plain text.	Med	PLAN
FR-5.6	Messages from a student who has been blocked by the viewer shall not be shown to that viewer.	Med	PLAN
FR-5.7	A member shall be able to leave a lobby that has not yet been confirmed; the freed place shall reopen the goal.	Med	PLAN

3.2.6 Module 6: Notes and Resource Sharing

Build Step 7 0 of 3 implemented

Private one-to-one responses to a notes request, with file attachments.

ID	Requirement	Pri.	Status
FR-6.1	A goal in the `NOTES` category shall be answered through a private one-to-one thread between the poster and each responder, not through a public lobby.	High	PLAN
FR-6.2	A responder shall be able to attach files (proposed: PDF and images, at most 10 MB each) that are kept in object storage and served by signed link.	Med	PLAN
FR-6.3	After receiving the material the poster shall be able to rate its quality separately from the person's reliability.	Low	PLAN

3.2.7 Module 7: Trust and Safety

Build Step 8 0 of 5 implemented

Reporting, blocking, administrator review and audited unmasking.

ID	Requirement	Pri.	Status
FR-7.1	A student shall be able to report another student or a goal with a reason; the report shall be stored for administrator review.	High	PLAN
FR-7.2	A student shall be able to block and unblock another student. Blocked students shall not see each other's goals, requests or messages.	High	PLAN
FR-7.3	Users shall have a role (`STUDENT` or `ADMIN`). The role column exists today; administrator-only routes are added with this module.	High	PART
FR-7.4	An administrator shall be able to list open reports and dismiss them, warn the reported student or suspend the account.	High	PLAN
FR-7.5	An administrator shall be able to unmask the poster of an anonymous goal only by giving a reason, and every unmasking shall be written to an `AuditLog`.	High	PLAN

3.2.8 Module 8: Reliability, Ratings and Feedback

Build Step 9 0 of 5 implemented

Attendance confirmation, no-show flags, ratings and the reliability score.

ID	Requirement	Pri.	Status
FR-8.1	After the goal's time, each lobby member shall confirm whether the meet-up took place. When the outcomes are recorded the lobby and the goal shall move to `COMPLETED`.	High	PLAN
FR-8.2	A member shall be able to flag a no-show against a specific member. The flag shall be recorded and counted in that student's `noShowCount`.	High	PLAN
FR-8.3	After completion a member shall be able to rate each other member from 1 to 5 with an optional comment, once per pair per goal.	Med	PLAN
FR-8.4	The reliability score shall start at 100, be recomputed on lifecycle events, and be shown on every profile and every feed card. Storage columns for the score and counters already exist.	High	PART
FR-8.5	`goalsCompleted` shall be increased for every member of a lobby that completes.	Med	PLAN

3.2.9 Module 9: Matching and Priority Queue

Build Step 10 0 of 4 implemented

Rule-based goal suggestions and reliability-aware request ordering.

ID	Requirement	Pri.	Status
FR-9.1	The feed shall be able to suggest goals ranked by shared interests, branch and categories the student has joined before.	Med	PLAN
FR-9.2	When several students request the same goal the poster's list shall be ordered by reliability and profile completeness, not only by arrival time.	Med	PLAN
FR-9.3	Hostel-block and proximity matching. Descoped: with a pilot confined to one batch, proximity had almost nothing to discriminate on.	Low	DESC
FR-9.4	Duplicate and near-duplicate goal detection. Out of scope for the course project, as stated in the proposal.	Low	DESC

3.2.10 Module 10: Metrics, Lifecycle and Operations

Build Steps 11 and 12 2 of 5 implemented

Archival sweep, evaluation metrics, CI and deployment.

ID	Requirement	Pri.	Status
FR-10.1	A student's feed shall hide expired goals at query time. A scheduled sweep shall additionally mark expired and finished goals `ARCHIVED`. The query-time part is built; the sweep is not.	Med	PART
FR-10.2	The system shall compute the evaluation metrics of Section 4.4 (lobby fulfilment, time-to-first-request, no-show rate, resource response rate) and show them to administrators.	Med	PLAN
FR-10.3	`GET /api/health` shall report service status, database reachability and uptime, answering `503` when the database is down.	Low	IMPL
FR-10.4	Every push shall run format, lint, type-check, test and build in CI (GitHub Actions).	Med	IMPL
FR-10.5	The frontend shall be deployed on Vercel and the API on Render or Railway, with PostgreSQL in production and SQLite in development.	Med	PLAN

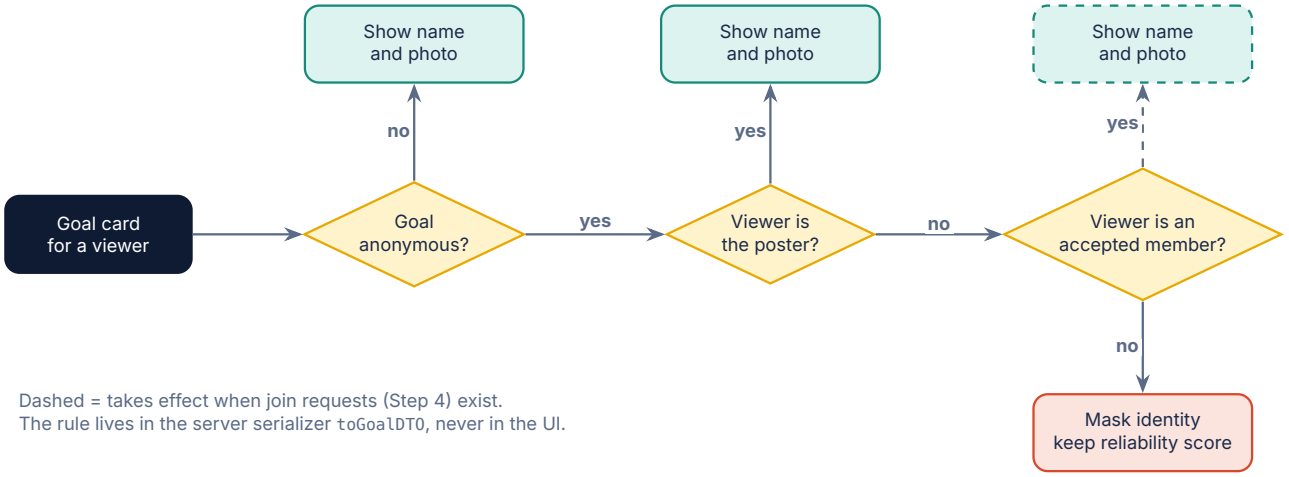


Figure 3.2. Decision applied to every goal card; the accepted-member branch becomes live with Step 4

3.3 Business Rules Behind the Requirements

3.3.1 Who can see a poster's identity

Anonymity is a pre-acceptance shield, not a permanent one. The decision is taken once, in the server-side serializer, for every goal returned to every viewer (Figure 3.2). The requester's own identity is never hidden from the poster, because the poster needs to see who is asking in order to approve.

3.3.2 Gender-preference matrix

A poster can restrict a goal to male or female requesters, for example for a late-night ride home. The rule below is the one proposed for Step 4; the preference itself is already stored and validated.

Preference	MALE	FEMALE	OTHER	UNDISCLOSED
`ANY`	allowed	allowed	allowed	allowed
`MALE_ONLY`	allowed	refused	refused	refused
`FEMALE_ONLY`	refused	allowed	refused	refused

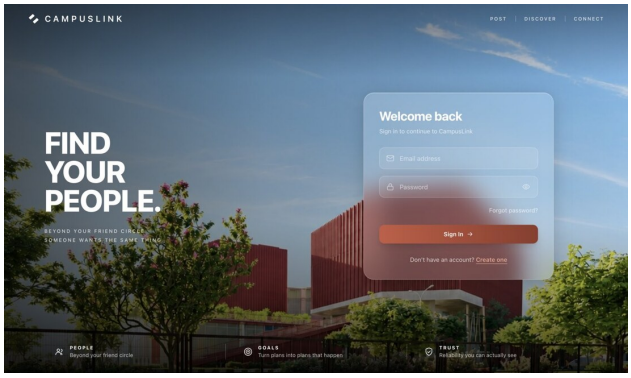
Table 3.1. Proposed enforcement of the gender preference

3.4 External Interface Requirements

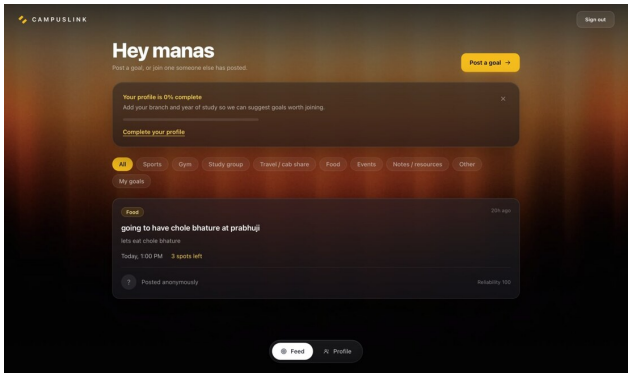
3.4.1 User interfaces

The interface is a responsive single-page application in a dark, high-contrast theme with a gold accent. A bottom navigation switches between Feed and Profile. Figure 3.3 shows three screens from the prototype.

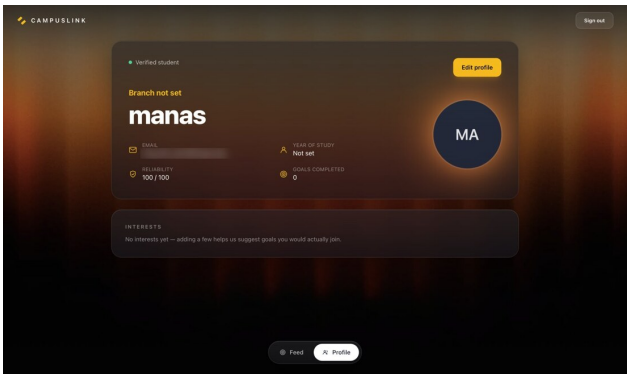
- Forms shall show validation messages next to the field and shall not clear the input on error.
- The feed card shall show category, title, time, remaining spots, poster (or "Posted anonymously") and reliability score.
- Interactive targets shall be at least 44 by 44 pixels on touch screens.



(a) Landing and sign-in



(b) Feed with category filters; one goal posted anonymously



(c) Profile with completeness and reliability

Figure 3.3. Prototype screens (e-mail address blurred)

3.4.2 Software interfaces: REST API

All routes are served under `/api`. Request bodies are JSON, validated with Zod; the server answers with the status codes `400`, `401`, `403`, `404`, `409`, `429` and `503` where applicable. The following 16 endpoints are implemented.

Method	Path	Access	Purpose
GET	`/health`	public	Service status, database reachability, uptime.
POST	`/auth/signup`	public, limited	Create an unverified account and e-mail an OTP.
POST	`/auth/resend-otp`	public, limited	Send a new OTP; identical response for unknown addresses.
POST	`/auth/verify-otp`	public, limited	Verify the code, mark the account verified, return a session.
POST	`/auth/login`	public, limited	E-mail and password; returns JWT and private user record.
GET	`/auth/me`	JWT	Current user from the token.
GET	`/users/me`	JWT	Own profile with completeness percentage.
PATCH	`/users/me`	JWT	Edit name, branch, year, gender, interests (strict schema).
POST	`/users/me/photo`	JWT	Upload an avatar (data URL; JPEG, PNG or WebP).
DELETE	`/users/me/photo`	JWT	Remove the avatar.
GET	`/users/:id/photo`	public	Avatar bytes, long-lived private cache.
GET	`/users/:id`	JWT	Public profile of another student (no e-mail, no role).
POST	`/goals`	JWT	Post a goal.

Method	Path	Access	Purpose
GET	/goals	JWT	Feed: filters category, from, to, mine; cursor, limit.
GET	/goals/:id	JWT	One goal, serialized for the viewer.
DELETE	/goals/:id	JWT, poster	Cancel an open goal.

Planned endpoints (proposed names) cover join requests (POST /goals/:id/requests, PATCH /requests/:id), lobbies and chat (GET /lobbies/:id, Socket.IO), reports and blocks (POST /reports, PUT /blocks/:userId), admin unmasking and metrics (/admin/*), and ratings (POST /ratings).

3.4.3 Hardware, communication and external-service interfaces

- **Hardware:** none beyond a device with a browser. The API needs no special hardware.
- **Communication:** HTTPS between browser and API in production; JSON over REST; WebSocket for chat (planned). CORS is limited to the front-end origin.
- **SMTP:** nodemailer sends the OTP e-mail through a configured relay (port 465 with TLS, or 587 with STARTTLS).
- **Object storage (planned):** Cloudinary or Firebase for notes attachments, reached from the server only.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

CampusLink is sized for a pilot of 15 to 20 students, so the targets below are modest. They are *proposed* and shall be measured in the pilot.

- **API latency:** the 95th percentile of feed, goal-detail and profile requests shall stay below 300 ms on the free-tier host.
- **Page load:** the single-page application shall become interactive in under 3 seconds on a typical campus Wi-Fi or 4G connection.
- **Feed query:** the feed shall be cursor-paginated (default 20, maximum 50 per page) and served by indexes on status with creation time, status with category, and date; query cost shall not grow with the number of old goals.
- **Chat delivery (planned):** a message shall reach every online lobby member within 1 second.
- **Payload limits:** JSON bodies are limited to 1 MB; avatars to 250 KB after resizing to 320 px.

4.2 Security Requirements

- **Credential storage:** passwords are stored only as salted scrypt hashes; OTP codes only as hashes, single-use, 10-minute expiry, 5 attempts.
- **Injection:** all database access goes through Prisma's parameterised queries; request bodies are validated with Zod before use.
- **Cross-site scripting:** React escapes output, and the front end contains no dangerouslySetInnerHTML. Chat text shall be rendered as plain text.
- **Access control:** every route except health, auth and the avatar image needs a valid Bearer token; only the poster may cancel a goal; administrator routes require role = ADMIN.

- **Mass assignment:** the profile update schema is strict and the service uses an explicit allow-list, so a student cannot set `role`, `isVerified` or `reliabilityScore`.
- **Enumeration:** resend-OTP and login give the same response for unknown and known addresses.
- **Abuse control:** authentication endpoints are limited to 20 requests per client per 15 minutes; `helmet` sets secure headers.
- **Anonymity:** enforced in the server serializer, never in the UI, so a hidden identity cannot be recovered from the API response.
- **Auditability (planned):** every administrator unmasking is written to an `AuditLog` with the reason.

Known security limitations

(1) The session token is kept in `localStorage`, so a cross-site-scripting flaw would expose it, and logout only discards it on the client. (2) E-mail aliases such as `name+tag@thapar.edu` are accepted as separate accounts, which lets a student re-register to shed a bad reliability score; the fix is a canonical-e-mail unique column and is deferred for the pilot. (3) Avatar images are served from an unauthenticated URL with a long private cache so that browsers can reuse them. They are not secret, but they are reachable by anyone who knows a user id.

4.3 Software Quality Attributes

- **Usability and responsiveness:** fluid layouts for phone and desktop; touch targets of at least 44 by 44 px; clear, specific error messages.
- **Reliability:** at least 99% availability during evaluation sessions; the client shall handle network errors without losing typed input.
- **Maintainability:** a monorepo with a layered back end (route, controller, service, serializer), one shared package for schemas and enumerations, and strict TypeScript. CI runs format, lint, type-check, test and build on every push.
- **Testability:** more than 90 automated tests cover auth, profile, goals, health, OTP, passwords, e-mail templates and profile completeness.
- **Portability:** the same code runs on SQLite in development and PostgreSQL in production through Prisma.
- **Scalability:** the API is stateless and the feed is paginated and indexed, so the design does not prevent horizontal scaling, even though a pilot does not need it.

4.4 Business Rules and Evaluation Metrics

4.4.1 Business rules in force

Rule	Value
E-mail domain	`thapar.edu`, stored lower-cased
Password	8 to 128 characters
OTP	6 digits; valid 10 minutes; 5 attempts; single use
Session	JWT, default lifetime 7 days (configurable)
Goal title and description	4 to 80 and 10 to 600 characters
Headcount	1 to 20, excluding the poster
Goal expiry	Dated goal: at its date and time. Undated goal: 48 hours after posting
Feed page	Default 20, maximum 50
Interests	Up to 10 tags of up to 24 characters
Avatar	JPEG, PNG or WebP; 320 px; at most 250 KB
Auth rate limit	20 requests per 15 minutes per client

Table 4.1. Constants fixed by the implementation

4.4.2 Reliability score

Every student starts at 100. The score is stored on the user and recomputed whenever a goal's outcome is recorded, because it is printed on every feed card and must not need a query per card. The following formula is *proposed* for Step 9:

$$R = \max\left(0,\ 100 - 10\,n_{\text{no-show}} - 5\,n_{\text{upheld}}\right) \tag{4.1}$$

where $n_{\text{no-show}}$ is the number of confirmed no-shows and n_{upheld} the number of reports upheld by an administrator. Ratings are shown next to the score and do not enter it.

4.4.3 Evaluation metrics

The primary metric tells whether the core loop works. The secondary metrics test the discovery and the trust features.

$$\text{Fulfilment rate} = \frac{\text{goals that reach their headcount before their deadline}}{\text{goals posted}} \times 100\% \tag{4.2}$$

$$T_{\text{first}} = t_{\text{first request}} - t_{\text{posted}} \tag{4.3}$$

$$\text{No-show rate} = \frac{\text{confirmed lobbies with at least one no-show}}{\text{confirmed lobbies}} \times 100\% \tag{4.4}$$

$$\text{Resource response rate} = \frac{\text{notes requests with at least one private response}}{\text{notes requests posted}} \times 100\% \tag{4.5}$$

The deadline of a goal is its own date and time, or 48 hours after posting for an undated goal. A short 1–5 question on perceived safety is asked after a student uses anonymous posting or the gender filter.

5 Other Requirements

5.1 Future System Evolution

- **Push notifications** on lobby activity.
- **Streaks and badges** for reliable students, and a **trending goals** section.
- **Canonical e-mail column** to close the alias loophole if the user base ever grows beyond a pilot.
- **Object storage for avatars,** moving them out of the database behind the same interface used for notes.
- **Password reset** through an e-mailed OTP (FR-1.12).

5.2 Implementation Roadmap

The build order is fixed by dependency. Steps 0 to 3 are complete; Step 4 is next. Figure 5.1 lists all thirteen steps with the requirement module each one delivers.

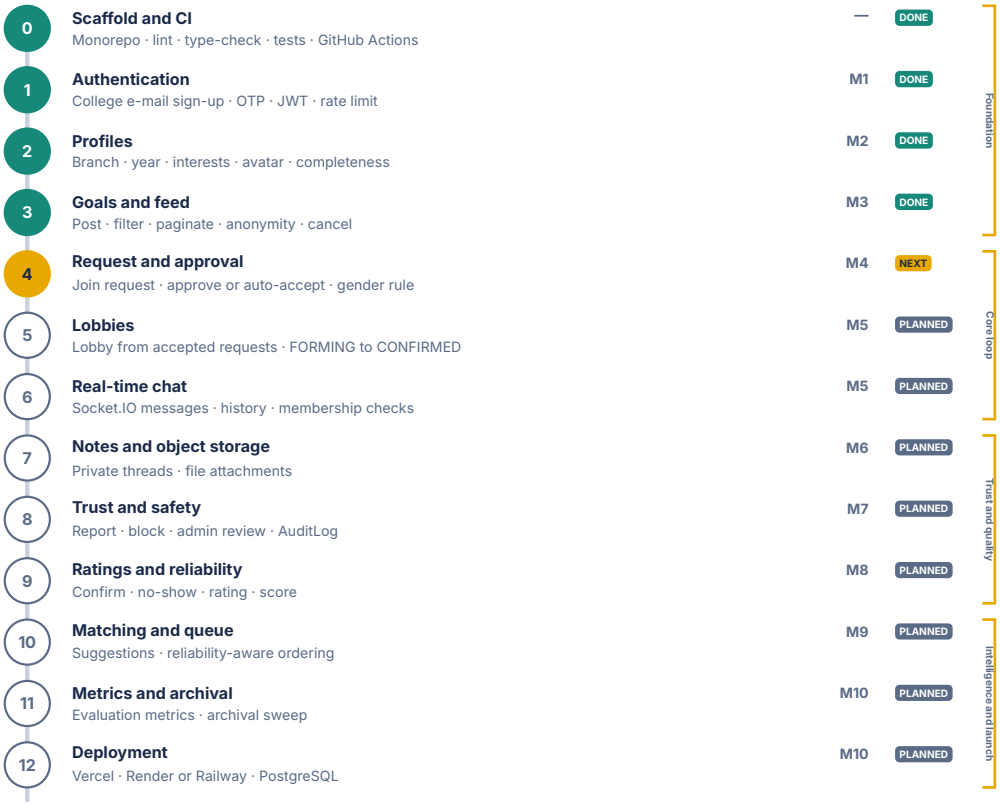


Figure 5.1. Build order of CampusLink; steps 0 to 3 are done, step 4 is next

5.3 Risks

Risk	Effect	Mitigation
Cold feed in the pilot	No goals, so request and lobby flows cannot be tested.	Seed a few goals alongside real ones.
Real-time chat on free tier	Sleeping instances delay the first WebSocket message.	Fall back to polling; keep the API stateless.
Alias accounts	A student sheds a bad score by re-registering.	Accepted for the pilot; canonical e-mail later.
Gender-rule edge cases	Students with undisclosed or other gender are refused on a restricted goal.	Document the rule; revisit after the pilot.
Mail delivery delay	Sign-up stalls while the passcode is awaited.	Resend endpoint and a clear message.

Table 5.1. Main project risks

A Assumptions and Open Points

These items were decided by the authors while writing the SRS and are collected here so that they can be confirmed or corrected.

1. **Gender rule.** A requester whose gender is `UNDISCLOSED` or `OTHER` is refused on a `MALE_ONLY` or `FEMALE_ONLY` goal (Table 3.1).
2. **Performance targets** in Section 4.1 (300 ms, 3 s, 1 s) are proposed, not measured.
3. **Planned endpoint names** in Section 3.4 are proposals.
4. **Reliability formula** in Section 4.4 is proposed; the proposal only requires that no-shows are factored in.
5. **Limits** of 1000 characters per chat message and 10 MB per attachment are proposed.
6. **`CLOSED` goal status** exists in the enumeration but has no defined transition.
7. **OTP at login.** The proposal could be read as asking for an OTP at every login. The implementation verifies once at sign-up, and this SRS follows the implementation.
8. **Lobby cancellation.** A lobby can be cancelled only while forming, because the implemented rule lets a poster cancel only an `OPEN` goal.
9. **Revision history** dates are approximate month-level entries.